

SOC Monitoring Home Lab using Splunk & Linux Security Logs

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Technologies: Splunk Enterprise

Windows Event Logs

Ubuntu Linux

SIEM

Security Monitoring

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Objective

The objective of this project was to build a Security Operations Center (SOC) Home Lab for monitoring, collecting, and analyzing security events from Windows and Linux systems. The project focused on log analysis, failed login detection, alert creation, and security monitoring using Splunk Enterprise and Linux authentication logs.

Environment Setup

Component	Details
Host OS	Windows 11
SIEM Tool	Splunk Enterprise
Linux VM	Ubuntu
Virtualization	VMware Workstation
Log Sources	Windows Security Logs, Linux Auth Logs

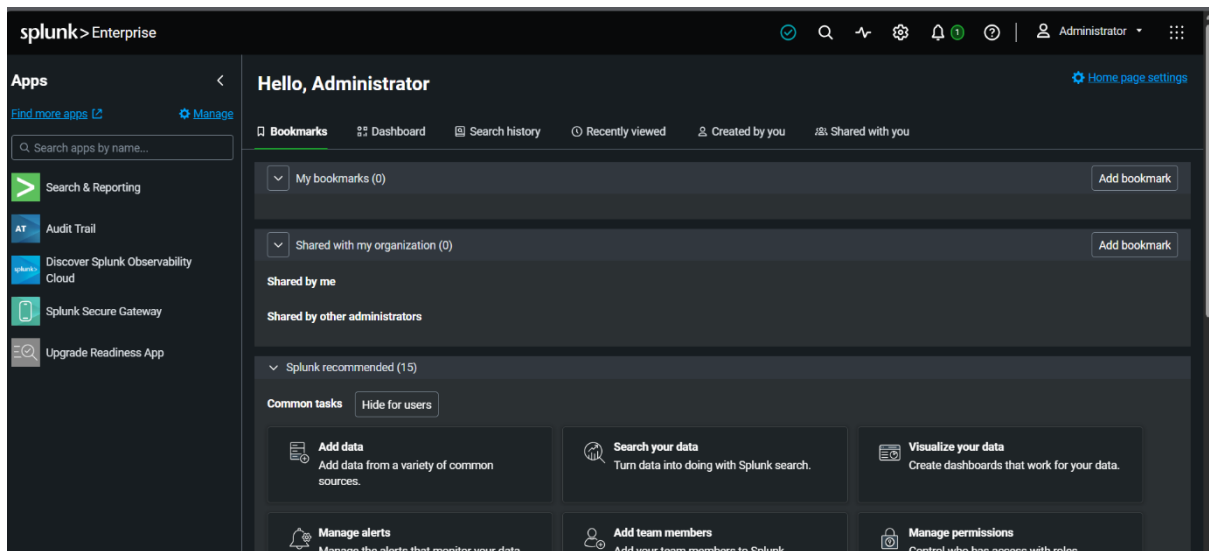


Fig 1 : Splunk Home Dashboard

```
sneha@ubuntu:~$ hostname
ubuntu
sneha@ubuntu:~$ whoami
sneha
sneha@ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:77:4c:6f brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.203.133/24 metric 100 brd 192.168.203.255 scope global dynamic ens33
        valid_lft 1672sec preferred_lft 1672sec
    inet6 fe80::20c:29ff:fe77:4c6f/64 scope link proto kernel_l1
        valid_lft forever preferred_lft forever
sneha@ubuntu:~$
```

Fig 2 : Ubuntu IP

Windows Security Monitoring

Windows Security, System, and Application logs were collected and monitored using Splunk Enterprise. Security events were indexed and made searchable for threat detection and investigation.

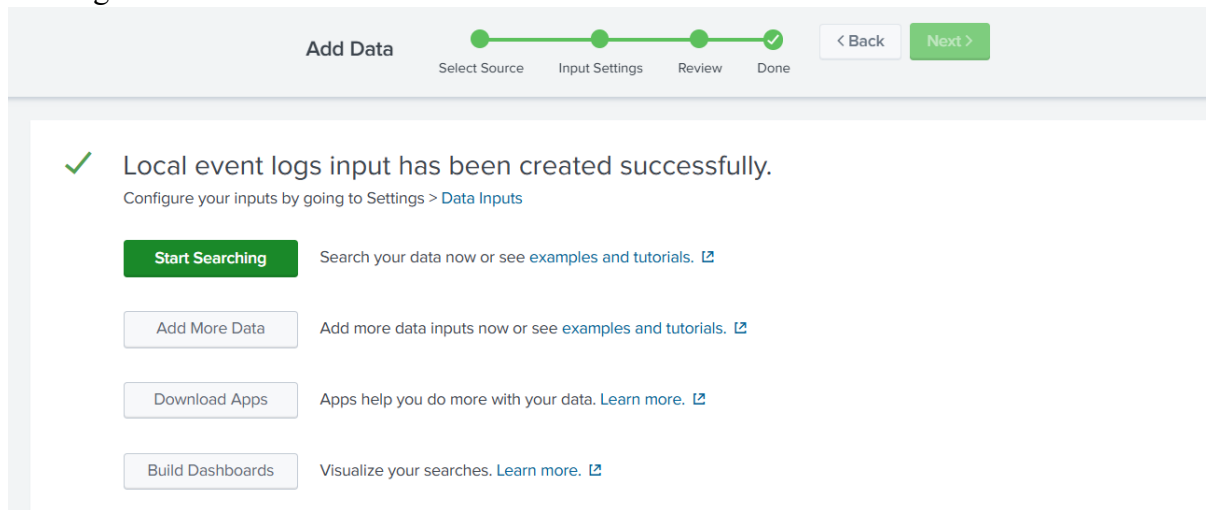


Fig 3 : Event Log Selection

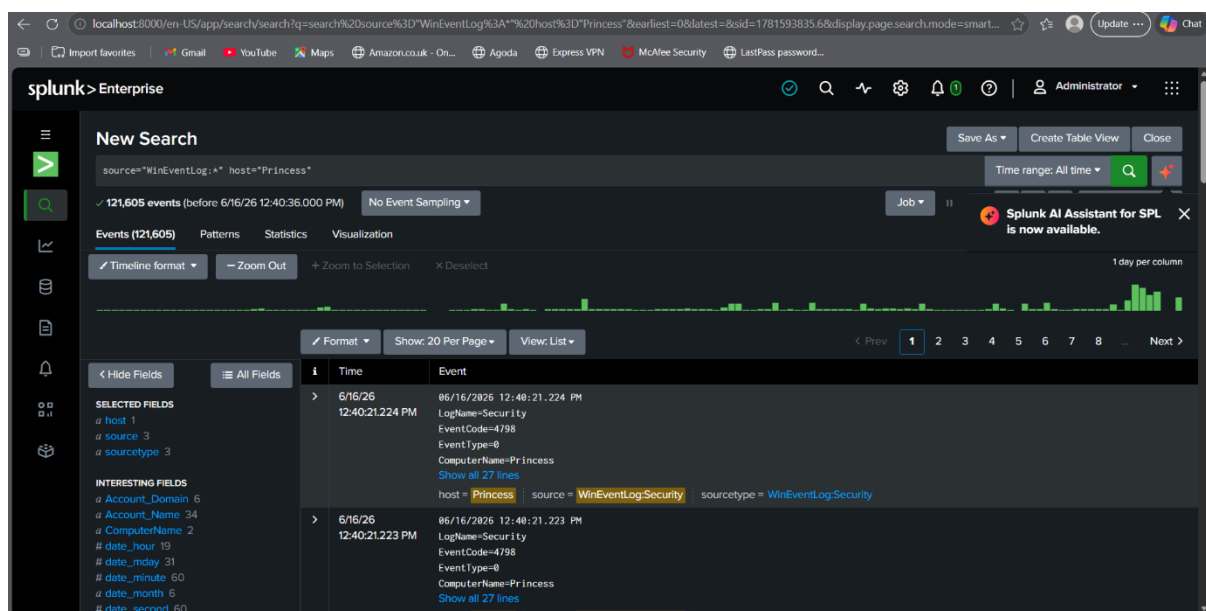


Fig 4 : First Ingested Logs

Failed Login Detection

Failed login attempts were intentionally generated and analyzed using Event ID 4625. These events simulate unauthorized access attempts and brute-force activity.

Query : `source="WinEventLog:Security" EventCode=4625`

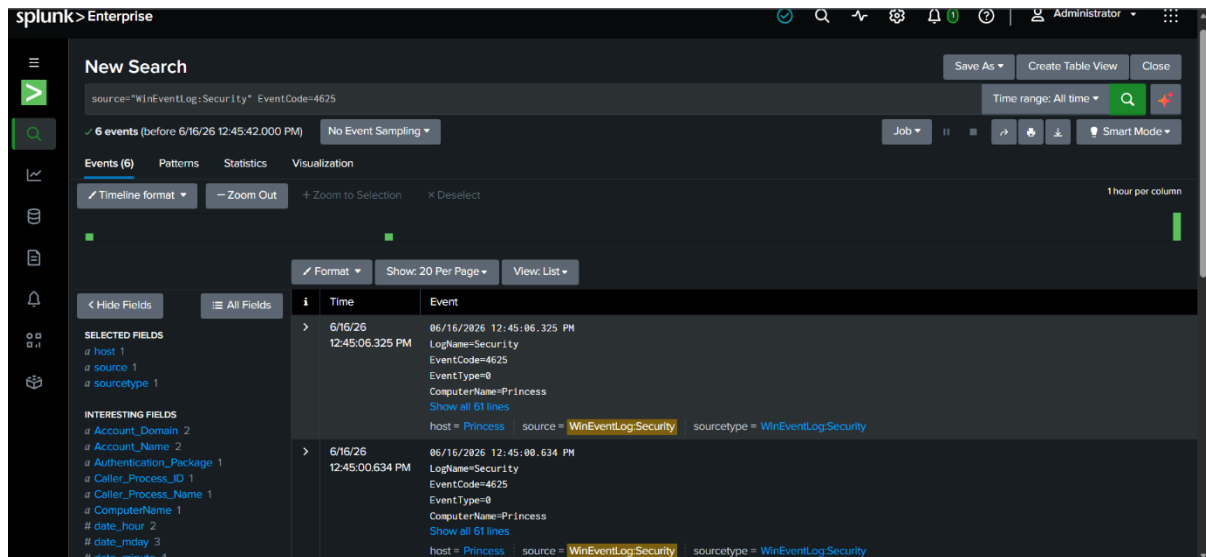


Fig 5 : Failed Login Detection

Alert Configuration

A Splunk alert was configured to automatically detect multiple failed login attempts within a defined time window.

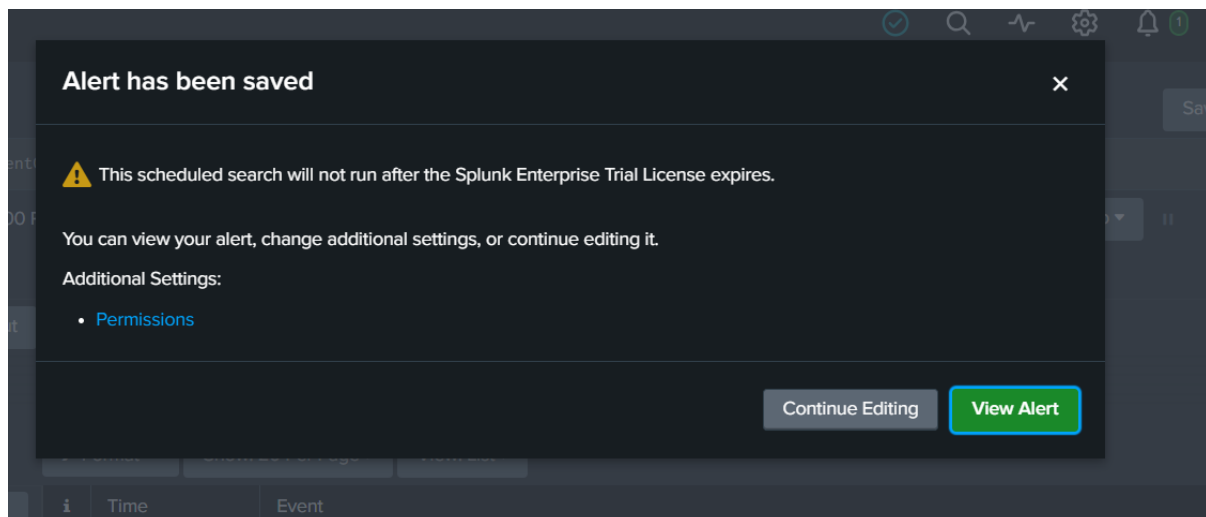


Fig 6 : Alert Created

Linux Security Monitoring

Linux authentication logs located in `/var/log/auth.log` were analyzed to investigate login activities, privilege escalation attempts, and authentication failures.

```
sneha@ubuntu:~$ sudo cat /var/log/auth.log | tail -20
[sudo: authenticat] Password:
2026-06-03T11:13:00.546411+00:00 ubuntu sudo: sneha : TTY=/dev/tty1 : PWD=/home/sneha : USER=root : COMMAND=/usr/bin/systemctl status elasticsearch
2026-06-03T11:13:05.586137+00:00 ubuntu sudo: pam_unix(sudo:session): session closed for user root
2026-06-03T11:17:01.565321+00:00 ubuntu CRON[2117]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-06-03T11:17:01.616193+00:00 ubuntu CRON[2117]: pam_unix(cron:session): session closed for user root
2026-06-03T11:21:34.361130+00:00 ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by sneha(uid=1000)
2026-06-03T11:21:34.373118+00:00 ubuntu sudo: sneha : TTY=/dev/tty1 : PWD=/home/sneha : USER=root : COMMAND=/usr/bin/systemctl status elasticsearch
2026-06-03T11:21:38.725788+00:00 ubuntu sudo: pam_unix(sudo:session): session closed for user root
2026-06-03T11:21:51.189792+00:00 ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by sneha(uid=1000)
2026-06-03T11:21:51.195733+00:00 ubuntu sudo: sneha : TTY=/dev/tty1 : PWD=/home/sneha : USER=root : COMMAND=/usr/bin/systemctl status kibana
2026-06-03T11:22:34.220442+00:00 ubuntu sudo: pam_unix(sudo:session): session closed for user root
2026-06-03T12:17:01.806832+00:00 ubuntu CRON[2268]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-06-03T12:17:01.839085+00:00 ubuntu CRON[2268]: pam_unix(cron:session): session closed for user root
2026-06-16T07:48:45.544069+00:00 ubuntu systemd-logind[1368]: Matching system buttons on /dev/input/event0 (Power Button)
2026-06-16T07:48:45.544077+00:00 ubuntu systemd-logind[1368]: Matching system buttons on /dev/input/event1 (AT Translated Set 2 keyboard)
2026-06-16T07:48:45.544085+00:00 ubuntu systemd-logind[1368]: New seat seat0.
2026-06-16T07:49:01.215262+00:00 ubuntu login: pam_unix(login:session): session opened for user sneha(uid=1000) by sneha(uid=0)
2026-06-16T07:49:01.271045+00:00 ubuntu systemd-logind[1368]: New session '1' of user 'sneha' with class 'user' and type 'tty'.
2026-06-16T07:49:01.403527+00:00 ubuntu (systemd): pam_unix(systemd-user:session): session opened for user sneha(uid=1000) by sneha(uid=0)
2026-06-16T07:49:01.408837+00:00 ubuntu systemd-logind[1368]: New session '2' of user 'sneha' with class 'manager' and type 'unspecified'.
2026-06-16T07:52:48.287091+00:00 ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by sneha(uid=1000)
sneha@ubuntu:~$ _
```

Fig 7 : Authentication Logs

```
sneha@ubuntu:~$ su root
Password:
su: Authentication failure
sneha@ubuntu:~$ su root
Password:
su: Authentication failure
sneha@ubuntu:~$ su root
Password:
su: Authentication failure
sneha@ubuntu:~$ su root
Password:
su: Authentication failure
sneha@ubuntu:~$ sudo cat /var/log/auth.log | grep -i fail | tail -20
2026-05-31T12:34:47.315503+00:00 ubuntu login: pam_unix(login:auth): authentication failure; logname= uid=0 euid=0 tty=/dev/tty1 ruser= rhost= user=sneha
2026-05-31T12:34:49.251290+00:00 ubuntu login: FAILED LOGIN 1 FROM tty1 FOR sneha, Authentication failure
2026-05-31T12:35:06.698294+00:00 ubuntu login: pam_unix(login:auth): authentication failure; logname= uid=0 euid=0 tty=/dev/tty1 ruser= rhost=
2026-05-31T12:35:09.703141+00:00 ubuntu login: FAILED LOGIN 2 FROM tty1 FOR su26, Authentication failure
2026-05-31T12:35:26.703337+00:00 ubuntu login: pam_unix(login:auth): authentication failure; logname= uid=0 euid=0 tty=/dev/tty1 ruser= rhost=
2026-05-31T12:35:29.841698+00:00 ubuntu login: FAILED LOGIN 1 FROM tty1 FOR su26, Authentication failure
2026-06-03T03:57:53.727773+00:00 ubuntu unix_chkpwd[2130]: password check failed for user (sneha)
2026-06-03T03:57:53.729015+00:00 ubuntu sudo: pam_unix(sudo:auth): authentication failure; logname=sneha uid=1000 euid=0 tty=/dev/tty1 ruser=sneha rhost= user=
sneha
2026-06-16T07:55:08.274728+00:00 ubuntu unix_chkpwd[2244]: password check failed for user (root)
2026-06-16T07:55:08.278945+00:00 ubuntu su: pam_unix(su:auth): authentication failure; logname=sneha uid=1000 euid=0 tty=/dev/tty1 ruser=sneha rhost= user=root
2026-06-16T07:55:10.427300+00:00 ubuntu su[2242]: FAILED SU (to root) sneha on tty1
2026-06-16T07:55:29.384468+00:00 ubuntu unix_chkpwd[2256]: password check failed for user (root)
2026-06-16T07:55:29.387517+00:00 ubuntu su: pam_unix(su:auth): authentication failure; logname=sneha uid=1000 euid=0 tty=/dev/tty1 ruser=sneha rhost= user=root
2026-06-16T07:55:31.712386+00:00 ubuntu su[2254]: FAILED SU (to root) sneha on tty1
2026-06-16T07:55:41.537055+00:00 ubuntu unix_chkpwd[2268]: password check failed for user (root)
2026-06-16T07:55:41.539176+00:00 ubuntu su: pam_unix(su:auth): authentication failure; logname=sneha uid=1000 euid=0 tty=/dev/tty1 ruser=sneha rhost= user=root
2026-06-16T07:55:43.549259+00:00 ubuntu su[2266]: FAILED SU (to root) sneha on tty1
2026-06-16T07:56:01.902274+00:00 ubuntu unix_chkpwd[2280]: password check failed for user (root)
2026-06-16T07:56:01.904356+00:00 ubuntu su: pam_unix(su:auth): authentication failure; logname=sneha uid=1000 euid=0 tty=/dev/tty1 ruser=sneha rhost= user=root
2026-06-16T07:56:03.647307+00:00 ubuntu su[2278]: FAILED SU (to root) sneha on tty1
sneha@ubuntu:~$
```

Fig 8 : Linux Failed Login

Key Findings:

- Successfully collected Windows Security, System, and Application logs.
- Detected failed Windows login attempts using Event ID 4625.
- Created automated security alerts.
- Investigated Linux authentication failures.
- Analyzed privilege escalation and user session events.
- Demonstrated cross-platform security monitoring capabilities.

Conclusion

The SOC Home Lab successfully demonstrated fundamental Security Operations Center activities including log collection, security monitoring, event analysis, alert creation, and threat detection across Windows and Linux systems. The project provided practical exposure to SIEM operations and real-world SOC workflows.